

1 The mitotic cell cycle occurs in plant cells and in animal cells. The cycle is a continuous process that can be divided into three main parts, interphase, mitosis and cytokinesis.

(a) One essential feature of the mitotic cell cycle is the replication of DNA in interphase, before mitosis begins.

(i) Explain why the replication of DNA needs to occur before mitosis begins.

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(ii) Name **two** enzymes that are essential for the successful replication of DNA in the S phase of interphase.

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(iii) The synthesis of histone proteins also occurs in interphase of the mitotic cell cycle.

State a phase of interphase during which the synthesis of histone proteins is likely to occur.

Explain your answer.

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(b) Figure 1.1 is a photomicrograph of part of a tissue where the cells are in different stages of the mitotic cell cycle.

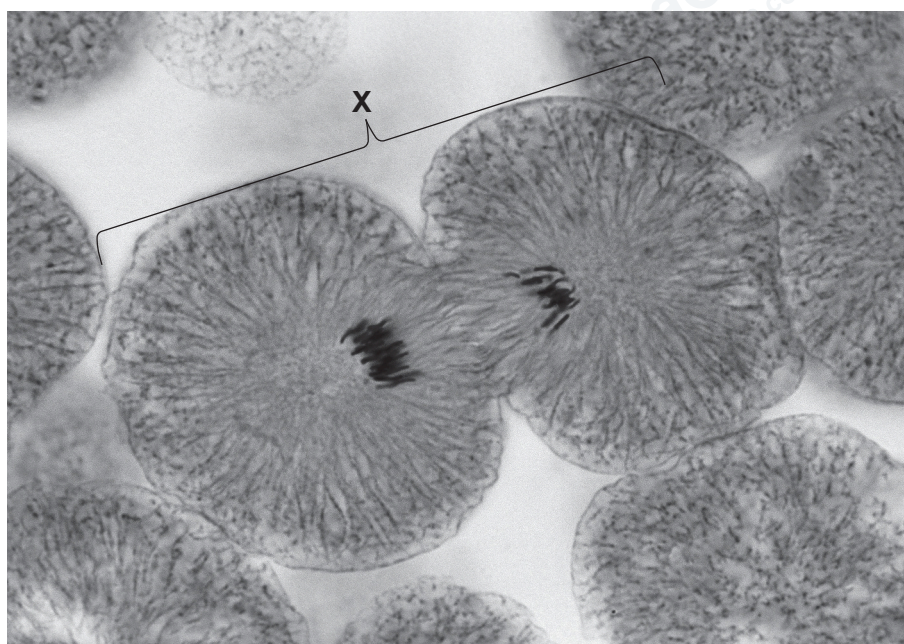


Figure 1.1

A student made two **incorrect** observations after studying Figure 1.1:

- Observation 1: Figure 1.1 shows plant cells.
- Observation 2: Area **X** shows two identical daughter cells in early interphase.

Explain why the observations made by the student are **not** correct.

Use evidence from Figure 1.1 to support your answer.

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[Total: 9]